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SUPERSEDING
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FEDERAL SPECIFICATION

GLOVES, LEATHER, WELDER'S, GAUNTLET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers horsehide and cattlehide work gloves used by welders, but does not include protective or safety gloves in which a rubber insert is used.

1.2 Classification.

1.2.1 *Types and sizes.* The gloves shall be of the following types and sizes as specified (see 6.2).

Type I—Welder's Gloves, Montpelier pattern (see figure 1).

Type II—Welder's Gloves, Gunn pattern (see figure 2).

Schedule of sizes

Small
Medium
Large
Extra-large

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276—Thread, Cotton.
KK-L-311—Leather; Methods of Sampling and Testing.
UU-P-268—Paper, Kraft, Untreated, Wrapping.
CCC-C-419—Cloth, Cotton, Duck, Unbleached, Plied—Yarns, (Army and Numbered).
CCC-T-191—Textile Test Methods.
DDD-L-20 — Label; For Clothing, Equipage, and Tentage, (General Use).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 751—Stitches, Seams and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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books and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-B-17757 — Boxes, Fiber, Corrugated (Modular Sizes).

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the supplier (see 6.3). Variation from the specification may appear in the sample in which case the specification shall govern.

3.2 Material.

3.2.1 *Leather.* The leather used shall be as specified in 3.8.1 and 3.9.1. The leather shall be chrome tanned, and pliable. When grain horsehide is used, it shall be full-grain horsehide fronts, i.e., sides with the butts trimmed off and the front shanks cut off. When split horsehide is used, it shall be flesh splits of horsehide fronts. When split cattlehide is used it shall be made from flesh split cattlehide. Middle splits, head and shoulder, or belly splits of horse or cattlehide shall not be used. Palms, thumbs, fingers, and backs shall be made from the best selected cuts.

3.2.1.1 *Thickness.* The thickness of leather glove parts shall be not less than 2-1/2 ounces nor more than 3-1/2 ounces.

3.2.1.2 *Color.* The color of all leather components shall be cream or light grey. All leather components used in each pair of gloves shall be of the same identical color.

3.2.1.3 *Physical requirements.*

3.2.1.3.1 *Stitch tearing strength.* Not less than 80.0 percent of the specimens tested shall have a stitch tearing strength of not less than 25 pounds when tested as specified in 4.2.1.1.

3.2.1.3.2 *Flexibility after soaking.* The leather shall not exceed its flex stiffness by more than 25.0 percent when tested as specified in 4.2.1.1.

3.2.1.4 *Chemical requirements.* The leather shall be fully chrome tanned and shall conform to the requirements of table I when tested as specified in 4.2.1.1.

TABLE I.—*Chemical requirements of leather*

	Minimum	Maximum
Chromic oxide, percent[1] ...	3.0	—
Chloroform extract, percent[1]	12.0	25.0
Ash, percent[1]	—	12.0
Acidity (pH)	3.5	5.0
Shrinkage temperature	97°C.	—

[1] Calculated on a moisture free basis.

3.2.2 *Thread cotton.* Thread for sewing the gloves shall be unbleached cotton thread conforming to type IA1 or IB1 of V-T-276. When stitch type 301 is used the thread shall conform to ticket No. 16, 4 ply. When stitch type 401 is used the top thread shall be ticket No. 16, 4 ply and the bottom thread shall be ticket No. 36, 4 ply.

3.2.3 *Knitted-wool/cotton cloth.* The back lining shall be a knitted wool/cotton plaited cloth. The face yarn shall be woolen spun yarn of 98.0 percent wool of not less than 50's grade (U.S. Standard) fleece or pulled wool. No reused or reprocessed wool or card waste or strippings shall be used. The backing yarn shall be a suitable cotton yarn and shall be not more than 30.0 percent of the weight of the cloth. The cloth shall be thoroughly scoured. The finished cloth shall conform to the following:

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Weight, ounces per square yard	9 to 11
Wales, per inch	17 to 20
Courses, per inch	17 minimum
Color	Unbleached, light grey, or tan

3.2.4 *Cotton duck.* The cotton duck cloth shall conform to type III of CCC-C-419 and shall have a weight of not less than 8.25 ounces per square yard.

3.2.5 *Label.* Each glove shall have a combination size and identification label conforming to type III or IV, class 4 of DDD-L-20. The marking medium shall be yellow or black and shall show fair colorfastness to accelerated laundering when tested as specified in DDD-L-20.

3.3 Stitches, seams, and stitching. All stitching and seaming shall be done with stitch type 301 or 401 and seam types as specified in table III except that 401 type shall not be used in the hemming at the opening of the gauntlet and shall be in conformance with the requirements of Federal Standard No. 751, using not less than 7 nor more than 10 stitches per inch. All ends of type 301 stitching shall be backstitched not less than 1/2 inch. All thread ends shall be trimmed (see 3.8.4 and 3.9.3).

3.4 Dies and patterns. The dies or patterns for cutting all parts of the gloves shall be furnished by the supplier and shall be of the proper proportions to provide good fitting gloves conforming to finished measurements in table II.

3.5 Figures. Figures 1 and 2 are furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figures, the written specification shall govern.

3.6 Dimensions of gloves. The finished gloves shall conform to the dimensions shown in table II. Measurements shall be taken as shown on figures 1 and 2.

3.7 Marking. The label specified in 3.2.5 shall be applied to the inside of the back of each glove at center, approximately 1/2 inch down from the finished top edge. The item description for use on the label shall be as follows:

GLOVES, LEATHER, WELDER'S, GAUNTLET, TYPE .....(I or II)

3.8 Type I welder's gloves, Montpelier pattern.

3.8.1 *Leather (components).*

3.8.1.1 *Front.* The palm, thumb, and fourchettes shall be of split-horsehide leather. The gauntlet piece shall be split-horsehide or cattlehide leather.

3.8.1.2 *Back.* The one-piece back shall be of grain-horsehide leather.

3.8.1.3 *Reinforcing welts.* The welts shall be of grain-horsehide or cattlehide leather.

3.8.1.4 *Dimensions of reinforcements.* The fourchette strips shall be approximately 3/4

TABLE II.—*Finished dimensions of gloves*

Measurement	Small	Medium	Large	Extra large	Tolerance, plus or minus
	Inches	*Inches*	*Inches*	*Inches*	*Inch*
A—Top edge or cuff to tip of second finger	12-3/4	13	13-1/2	14	1/2
B—Bottom edge of cuff to tip of second finger	7-3/4	8	8-1/2	8-3/4	1/4
C—Across palm at thumb crotch, seam to seam	4-1/2	5	5-1/2	5-3/4	1/8
D—Tip of index finger to thumb crotch	4-3/4	5	5-1/2	5-3/4	1/8
E—Thumb crotch to tip of thumb	2-3/4	3	3-1/4	3-1/2	1/8
F—Across bottom edge of cuff	4-3/4	5	5-1/2	6	1/4
G—Across top edge of cuff	6-1/4	6-3/4	7	7-1/4	1/4

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inch wide. The welt strips for reinforcing seams shall be cut approximately 7/32 inch wide.

3.8.2 *Design.* Type I gloves shall be of the pattern known commercially as Montpelier, as shown in figure 1. The gloves shall be made of two sections with finger fourchettes. The front section shall be composed of a separate gauntlet piece, a face of thumb-palm-finger piece, and a back of the thumb piece. The back section shall be made in one piece with the gauntlet, back, and fingers continuous. The gauntlets of both the front and the back shall be lined with the cotton duck specified in 3.2.4. In addition, the back section shall be lined from the bottom edge of the gauntlet duck lining to the ends of the fingers with the knitted cloth specified in 3.2.3. The palm, thumb, and fingers of the front section shall be unlined.

3.8.3 *Construction.* The gauntlet piece shall be lined with the cotton duck to within 3/8 inch of the top edge of the gauntlet to facilitate the hemming of the top edge, and the gauntlet and duck joined to the palm and thumb by "inseam" construction with a protective welt strip inserted and caught in the closing seaming. The back of the glove shall be made in one continuous piece of leather and lined throughout; the gauntlet portion shall be lined with cotton duck specified in 3.2.4, and the remaining area with the knitted cloth, specified in 3.2.3, lapped under the bottom edge of the duck lining and extending to the ends of the fingers. The thumb shall be joined to the palm by "inseam" construction with an "all-around" welt strip inserted and caught in the seaming. The fronts shall be joined to backs by "inseam" construction with a welt strip caught in the seaming from the top of the glove to the top of the little finger and the seaming continued around the fingers. The closing shall be of "inseam" construction with a protective welt strip extending from the top of the first finger, inserted and caught in the seaming. Welts shall protrude from seams not more than 1/8 inch. The back lining shall be assembled into the glove with the cotton back next to the leather and the wool fleece face exposed. The back lining shall be caught in finger and closing seams. The ends of the welt shall extend to within 1/2 inch of the top of the finished glove and shall not be caught in the hem. The fourchettes shall be joined to the fronts and the backs by inseam construction.

3.8.4 *Stitches and seams.* Stitches and seams of type I gloves shall conform to the requirements of table III and 3.3.

TABLE III.—*Stitches and seams in type I welder's gloves, Montpelier pattern*

Operation	Gage	Seam type
Thumb and closing seams	—	SSj-1
Back lining to gauntlet lining	1/8 to 3/16	LSa-2
Palm and back of thumb to gauntlet	—	SSj-1
Gauntlet hem	—	EFa-1
Fourchettes	—	SSa-1

3.9 Type II welder's gloves, Gunn pattern.

3.9.1 *Leather.* Type II gloves shall be of split-cattlehide leather; welts, shall be split cattlehide and shall be cut approximately 7/32 inch wide.

3.9.2 *Design.* Type II gloves shall be of the pattern known commercially as Gunn-cut, as shown in figure 2. The face of the thumb, the palm, and the first (index) and fourth (little) fingers shall be made of one piece of leather. The back shall be of one piece and shall include the backs of the fingers and gauntlet. The palm sides of the second and third fingers shall be of one piece. The back of the thumb shall be of one piece. The gauntlet front shall be of one piece. The gauntlet cuffs (front and back) shall be fully lined or faced with cotton duck specified in 3.2.4. The facing may be of one or two pieces. The front and back of the facings shall be cut shorter than the outside of the cuffs, and the outside of the cuffs shall be turned over the top and

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stitched to form a self-binding. There shall be a lining of the knitted cloth specified in 3.2.3 in the back extending from the bottom of the gauntlet facing to the tips of the fingers. The back lining shall be assembled into the glove with the cotton back next to the leather and the wool fleece exposed.

3.9.3 *Stitches and seams.* All seams with the exception of the seams at the base of second and third fingers, back lining to gauntlet lining, palm and back of thumb to gauntlet, and the hemming shall be leather welted and welts shall protrude from the seams not more than 1/8 inch. The ends of the welts shall extend to within 1/2 inch of the top of the finished glove and shall not be caught in the hem. The seam at base of second and third fingers shall be covered with a leather flap. The flap shall be cut sufficiently wide so that when turned back into position the free portion shall be not less than 1/4 inch nor more than 1/2 inch in width. The ends of the flap shall be 3/8 ± 1/8 inch from the closing seam. The top edge of back lining shall be lapped under the bottom edge of gauntlet lining. Back knitted lining shall be caught in finger and closing seams. Stitches and seams shall conform to table IV and 3.3.

TABLE IV.—*Stitches and seams in type II gloves, Gunn pattern*

Operation	Gage	Seam type
Thumb, finger, and closing seams	—	SSj-1
Base of 2d and 3d fingers to palm	1/16 to 1/8	LSa-2
Back lining to gauntlet lining	1/8 to 3/16	LSa-2
Palm and back of thumb to gauntlet	1/8 to 3/16	LSa-2
Gauntlet hem	—	EFa-1

3.10 Pairing. The gloves shall be matched, paired and tacked together with thread. Tacking shall be done with not less than two stitches using a cord thread just below the hem approximately 1/2 inch from the closing seam, which will permit the pair to be pulled apart without tearing the leather.

3.11 Workmanship. The finished gloves shall conform to the quality and grade established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Inspection shall be in accordance with the provisions of MIL-STD-105 except where otherwise indicated herein. In addition to sampling inspection, as set forth in MIL-STD-105, process inspection will be performed during manufacture of the gloves to determine compliance with all operations and processes which cannot be determined in the completely fabricated gloves. Any deviation found in process inspection may be cause for rejection of the end item or items.

4.2.1 *Inspection of components.* In addition to the quality assurance provisions of the applicable specifications and drawings, components and materials shall be tested as specified in 4.2.1.1 and 4.2.1.2.

4.2.1.1 *Leather testing.* The leather shall be tested for the characteristics in table V. The test methods listed in table V shall be in accordance with KK-L-311, wherever applicable. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with the sampling of non-fabricated leather in Sampling and Acceptance Procedures of KK-L-311. When the

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TABLE V.—*Leather component tests*

Component and lot size expressed in terms of:	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	Number of determinations per sample unit	Results reported as:	AQL
Leather (when horsehide—1 front) (when cattlehide 1 side)	Material identification	3.2.1	Std. comm'l	X	—	1	Pass or fail	6.5
	Tannage	3.2.1	Std. comm'l	X	—	1	Pass or fail	6.5
	Stitch tearing strength	3.2.1.3.1	2151[1]	—	X	—	—	N/A
	Flexibility after soaking	3.2.1.3.2	4.3.1	X	—	1	Pass or fail	6.5
	Chromic oxide	3.2.1.4	6515	—	X	—	—	N/A
	Chloroform extract	3.2.1.4	6311	—	X	—	—	N/A
	Ash, total	3.2.1.4	6421	—	X	—	—	N/A
	Acidity (pH)	3.2.1.4	6621	—	X	—	—	N/A
	Shrinkage temperature	3.2.1.4	7011[2]	—	X	—	—	N/A

[1] Exception to method 2151. Thickness of the specimen need not be determined.
[2] Exception to method 7011. The heating medium shall be water.

data in the "Number of determinations per sample unit" and the "Results reported as" columns are not specified in table V, they shall be as required by the referenced test method.

4.2.1.2 *Cloth testing.* The cloth shall be tested for the characteristics in table VI. The test methods listed in table VI shall be in accordance with CCC-T-191, wherever applicable. Testing for all requirements listed in table VI is applicable to the sample unit. The sample unit for all required tests shall be 1/2 yard full width. The sample size for each characteristic shall be as follows:

Lot size (yards)	*Sample size*
800 or less	2
801 to 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more units fail to meet any requirements specified.

TABLE VI.—*Cloth component tests*

Component	Characteristic	Requirement reference	Test method
Knitted cloth (backlining)	Identification of face yarn	3.2.3	1100
	Identification of backing yarn	3.2.3	1200
	Percent of cotton in cloth	3.2.3	2100[1]
	Weight of cloth	3.2.3	5041
	Wales per inch	3.2.3	5070
	Courses per inch	3.2.3	5070

[1] 100 minus percent of wool equals percent of cotton backing yarn in cloth.

4.2.2 *Inspection of the end item.*

4.2.2.1 *Examination of the end item.* The defects found during examination of the end product shall be classified in accordance with 4.2.2.1.1 and 4.2.2.1.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.2.2.1.3. The sample unit for this examination shall be one glove. Selection of the sample shall be

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by pairs. The sample unit for pairing defects shall be one pair. Defects for pairing shall be classified as a single defect. The lot size shall be expressed in terms of one glove each.

4.2.2.1.1 *Visual examination of gloves.* The gloves shall be examined for defects in pairing, color, design, construction, material, workmanship and marking.

Examine	Defect	Classification	
		Major	Minor
Pairing	Not properly mated, e.g., not right and left of same size.	X	
	Not tacked together.		X
	Wide variation in color in the pair.		X
Color	Not as specified.	X	
	Color not uniform or shaded parts clearly noticeable.		X
Design	Any characteristic not in accordance with the specified requirements (unless otherwise classified herein).	X	
Cleanliness	Spots or stains, clearly noticeable.		X
Material (Leather)	Any through hole, cut, tear or rip.	X	
	Not cattlehide or horsehide (where specified).	X	
	Not full grain (where specified).	X	
	Not split leather (where specified).		X
	Cattlehide leather is cut from head, shoulder or belly stock.	X	
	Leather is middle split.	X	
	Loose, spongy or flanky.	X	
	Not soft and pliable.	X	
	Open vein.		X
	When applicable, grain damage penetrating deeply into leather fiber.	X	
	When applicable, grain damage not penetrating deeply into leather fiber.		X
	Any open scratch or grub hole.	X	
	Scratch or grub or tick marks not penetrating deeply into leather fiber.		X
	Deep fat wrinkles.	X	
	Any flesh cut affecting serviceability or appearance.	X	
	Any scar, brittle or bony area, weak or thin spot, or abraded area.	X	
	Healed grub marks or other healed damage affecting serviceability or appearance.	X	
	Flesh side of grain leather or either side of split leather not smooth or contains excessively loose flesh more than two square inches in total area.		X
(Knitted back lining or gauntlet lining)	Not type specified.		X
	Any hole, cut, tear, rip or drop stitch.	X	
	Multiple floats, loose ends, smashes, or slubs.		X
	Any broken or missing pick, end, course, or wale.		X
	Any mend in cloth (except restitching of seams).		X
Construction and workmanship, general (applicable to all components unless otherwise indicated herein)	Any component or operation omitted.	X	
	Any component misplaced not serving purposes intended.		X
	Any component or assembly damaged.	X	
	Any component not securely affixed, e.g., gauntlet loosely attached to back or palm of glove.	X	
	Any workmanship damage, e.g., scissor or knife cut.	X	

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Examine	Defect	Classification	
		Major	Minor
Construction and workmanship, general (applicable to all components unless otherwise indicated herein) (cont'd)	Any mend in leather, e.g., patch, (not applicable to restitched seam repair).	X	
	Needle chews.	X	
Seams and stitching	Any open seam. NOTE: When one or more stitches are broken or two or more consecutive skipped stitches or run-offs occur on a single stitched seam or on both rows of stitching or a double stitched seam, it shall be classified a *major* defect. When one or more stitches are broken or two or more consecutive skipped stitches or run-offs occur on only one row of a double stitched seam, it shall be classified as a *minor* defect.		
	Not specified seam type.	X	
	Not specified stitch type.	X	
	Any row of stitching omitted.	X	
	Loose stitch tension resulting in a loosely secured seam.	X	
	Tight stitch tension resulting in cutting of leather or breaking of stitches when normal pull is applied.	X	
	Ends of stitching not backstitched as specified.		X
Seams and stitching (cont'd)	Gage of stitching irregular or not as specified seriously affecting serviceability.	X	
	Gage of stitching irregular or not as specified affecting serviceability but not seriously.		X
	Any part caught in unrelated row of stitching.	X	
	One or two stitches per inch less than specified.		X
	Three or more stitches per inch less than specified.	X	
	More than the specified maximum number of stitches per inch damaging leather.	X	
	More than the specified maximum number of stitches per inch not damaging leather.		X
Assembly detail	Glove not neatly laid off.		X
	Finger distorted or twisted affecting comfort of wearer.	X	
	Thread ends not trimmed throughout glove.		X
	Not properly trimmed or finished, e.g., lump or bulk in finger inseam because leather was not properly trimmed.	X	
	Not constructed with specified number of pieces of leather (except welting).	X	
	Welting or reinforcing flap, if required, missing, or not caught by stitching.	X	

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Examine	Defect	Classification	
		Major	Minor
Assembly detail (cont'd)	Any welting strip end that finishes more than 5/8 inch from top edge of cuff.		X
	Gauntlet not fully faced or lined, or glove not lined where specified.	X	
Marking	Missing, incomplete, incorrect, not legible, not specified size, not in proper location or not accomplished in the specified manner.		X

4.2.2.1.2 *Dimensional examination.* The gloves shall be examined for dimensional defects including thickness of leather. Any dimension that is not within the established tolerance shall be classified a defect.

NOTE: Each leather part of the glove shall be checked for thickness, i.e., thumb, body of gloves, and fourchette strips. Any of these parts that are not within the required thickness range shall constitute a defect. (Thickness shall be determined in accordance with the procedure for "Fabricated-Leather Articles" of method 1011 of KK-L-311.) Where glove parts are lined, the lining should be parted from the leather and the thickness of the leather should be taken doubled. This dimension is then divided by two.

4.2.2.1.3 *Inspection levels and acceptable quality levels (AQL's).* The inspection levels and acceptable quality levels, expressed in defects per 100 units shall be as follows:

Examination	Inspection level	AQL	
		Major	*Total*
For defects in 4.2.2.1.1	II	2.5	6.5
For defects in 4.2.2.1.2	L6	—	4.0

4.2.3 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L4 and the AQL shall be 4.0 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Content	Number per container is more or less than required.

4.3 Tests.

4.3.1 *Test for flexibility after soaking.* The specimen shall be 3 by 4 inches, cut with the long dimension perpendicular to the backbone. The specimen shall be folded in half, grain side out, to form a rectangle 3 by 2 inches (crease parallel to backbone), then carefully folded in half again to form a rectangle 3 by 1 (approximate) inch, 4 layers thick. The folded specimen shall be secured with staples across the width, 1/4 inch from each end. Each of the 15 specimens shall be marked for ready identification for determining the percent increase in flex stiffness of the specimen after soaking as compared with the same specimen before soaking. The ends shall be gripped between thumb and forefinger and the specimens shall be twisted torquewise 40 times by describing an angle of 180 degrees. One end of the specimen shall then be fastened in the vise of the tester in such a way that the cut edges on the upper side face outward from the machine. The other end

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of the specimen shall extend 2 inches from the vise. The stiffness of the specimen shall be determined as specified in method 4211 of KK-L-311 using a 1 inch span and a bending moment of 0.5 inch-pound. The results shall be recorded in pounds. The folded stapled specimens shall then be immersed in distilled water of 140° ± 5°F. (60° ± 3°C.) for 20 ± 2 hours; 1,000 ml. of water shall be used for 15 specimens. Complete immersion of the specimens shall be accomplished by placing a floating cover on the vessel used for the immersion. The specimens shall be removed, air-dried, conditioned for 72 hours and retested for stiffness as described for unsoaked specimens. The percent increase in flex stiffness shall be calculated as follows:

$$\frac{S_2 - S_1}{S_1} \times 100$$

S_2 is the stiffness, in pounds of the soaked specimen.

S_1 is the stiffness, in pounds, of the unsoaked specimen.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing protection for civil agency procurements.)

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Sixty pairs of gloves, of one type and size only, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to class weather-resistant, size 3A of MIL-B-17757. The gloves shall be arranged flat 4 pairs in length (alternated fingertip to cuff), 1 pair in width, and 15 pairs in depth within a shipping container. Every other pair of gloves in each stack shall also be alternated fingertip to cuff. When applicable, the contents shall be immobilized by filling the voids with fiberboard filler pads or crumpled kraft paper. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268.

5.1.2 *Level B Military agencies.* Sixty pairs of gloves, of one type and size only, shall be packed in a fiberboard shipping container assembled and closed conforming to class domestic, size 3A of MIL-B-17757. The gloves shall be arranged flat 4 pairs in length (alternated fingertip to cuff), 1 pair in width, and 15 pairs in depth within a shipping container. Every other pair of gloves in each stack shall also be alternated fingertip to cuff. When applicable, the contents shall be immoblized by filling the voids with fiberboard filler pads or crumpled kraft paper. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268.

5.1.2.1 *Level B (civil agencies).* Seventy-two pairs of small, medium or large gloves of one type only; or sixty pairs of X-large gloves, of one type only, shall be packed in a fiberboard shipping container assembled, and closed conforming to class domestic, size 3A of MIL-B-17757. Small, medium, and large gloves shall be arranged flat 4 pairs in length (alternated fingertip to cuff), 1 pair in width, and 18 pairs in depth within a shipping container. Extra large gloves shall be arranged flat 4 pairs in length (alternated fingertip to cuff), 1 pair in width, and 15 pairs in depth within a shipping container. Every other pair of gloves in each stack shall also be alternated fingertip to cuff. When applicable, the contents shall be immoblized by filling the voids with fiberboard filler pads or crumpled kraft paper. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268.

5.1.3 *Level C.* Gloves shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest

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transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.1.4 *Standard Pack (Civil agencies).* The standard pack for this commodity shall be as specified in 5.1.2.1.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Gloves made in accordance with this specification are for use by welders and personnel engaged in similar occupations, as authorized. The type II welder's gloves (Gunn pattern) shall be used by all of the Military Services.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Type and sizes required (see 1.2).
(*c*) Selection of applicable level of packing (see 5.1).

6.3 Samples. For access to guide sample address the procuring activity issuing the invitation for bids.

6.4 Standard pack (civilian agencies). The standard pack requirements in paragraph 5.1.2.1 are intended for use in procurement of stores stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue, or shipment to eventual user.

6.5 Transportation description. The transportation description and minimum weights applicable to the commodity are:

Rail:

Gloves, not otherwise indexed by name.
Carload minimum weight 20,000 pounds, subject to rule 34, Uniform Freight Classification

Motor:

Gloves not otherwise indexed.
Truckload minimum weight 12,000 pounds, subject to rule 115, National Motor Freight Classification

MILITARY INTERESTS:

Army—GL

Navy—S&A

Air Force—MAA

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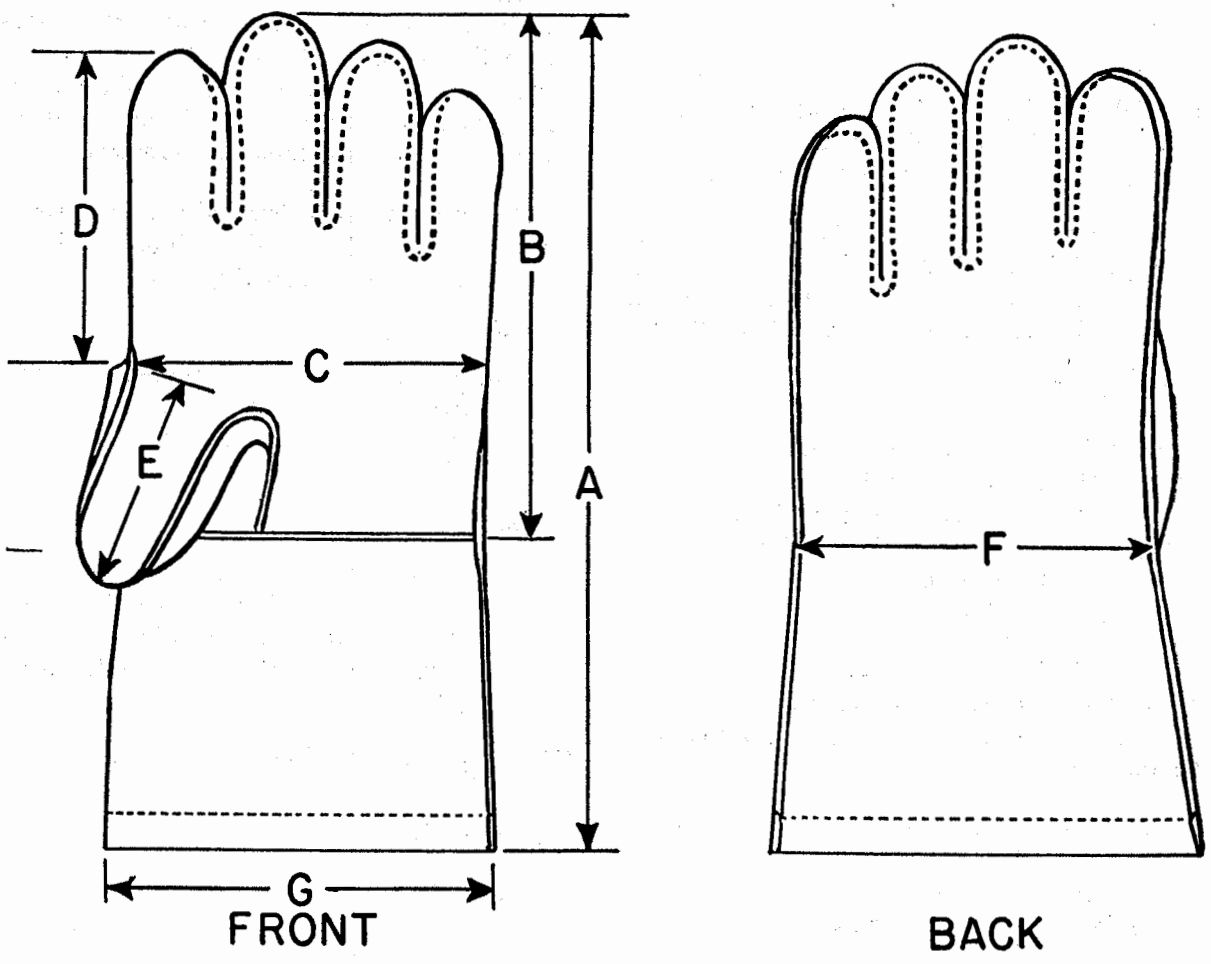


FIGURE 1.—Type I, welders' gloves, Montpelier pattern.

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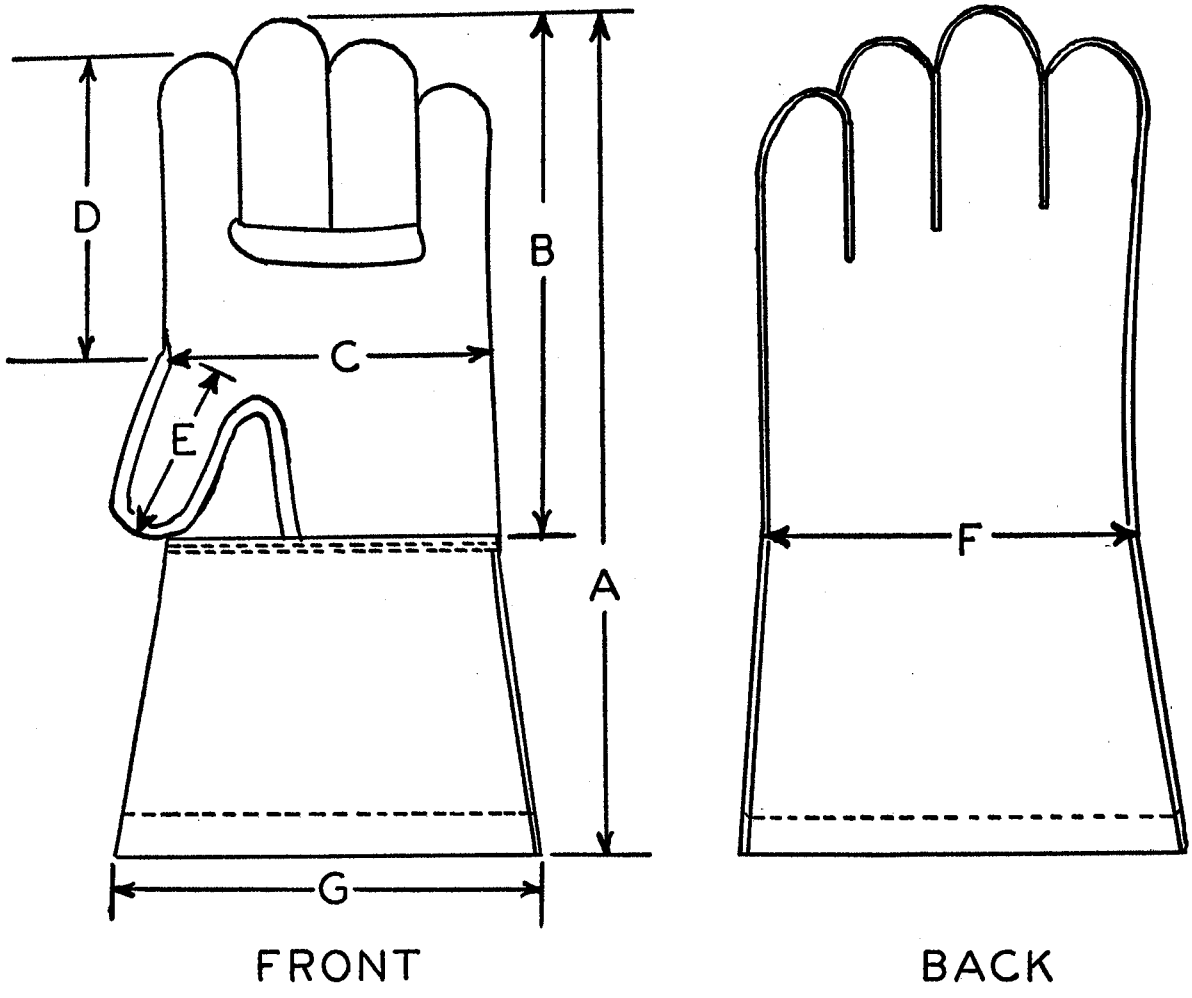


FIGURE 2.—Type II, welders' gloves, Gunn pattern.

Copies of this specification may be purchased for cents each.

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